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*Editorial*

# Review of Construction Technology of Advanced Energy Infrastructure

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**Abstract:** Energy is crucial to the development of human civilization. Energy infrastructure, such as oil and gas pipelines, power generation systems, and storage facilities, provide core support for the exploitation and utilization of various types of energy. Thus, energy infrastructure is vital to the economic sustainable development of a country. This paper provides the motivations and a brief introduction to the Special Issue entitled "Frontiers in Construction Technology of Advanced Energy Infrastructure", which aims to present advanced technologies and theories for energy infrastructure. The primary challenges in the current construction technology of energy infrastructure are described. Furthermore, the focus of current research in this field addressed in this Special Issue is presented. A comparison of the articles included or considered for inclusion in this Special Issue with other available literature on this topic is performed, which proves the prospects and relevance of this Special Issue. Finally, perspectives on future directions of energy utilization and energy infrastructure construction are provided.

**Keywords:** energy infrastructure; energy utilization; construction technologies; storage facilities; emerging marine industries; technological challenges

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## 1. Introduction

Energy infrastructure pertains to the comprehensive systems, structures, and facilities explicitly designed to produce, transmit, and distribute energy [1–3]. Serving as a cornerstone of modern societies, energy infrastructure powers homes, industries, and businesses [4,5]. It manifests in a variety of forms, depending on the specific energy source and other elements of its value chain, for instance, traditional energy infrastructure like gas pipelines, power generation systems for both fossil and renewable fuels, electrical transmission lines which are called lifelines, metering and distribution systems, intelligent systems, and modern electric and electronic systems [6,7]. It also includes storage facilities and different power control systems [8,9]. To expand further, present-day energy infrastructure takes the form of industrial systems related either directly or indirectly, and multiple interconnected sectors, like the transportation industry (charging units for electric vehicles, petrol station) and agricultural industry (farm machinery, drying systems) [10,11]. At a macro-level, energy infrastructure signifies a complex, multi-faceted system integral to the seamless functioning of modern societies. It serves as a pivotal engine propelling global economic growth and competitiveness [12–14].

Thus, energy development and utilization systems are fundamental cornerstones for the overall development of economies across the globe. Energy infrastructure is essential for effective resource usage, the normal function of a city, and waste energy disposal, but also indirectly affects economic and social development [15,16]. Over the past few decades, the focus of research pertaining to energy infrastructure has undeniably varied, carving a novel path in response to societal needs and technological advancements [17,18].

Bernd Resch et al. [19] analyzed the shortcomings of previous approaches to using GIS in renewable energy-related projects, extracted different challenges from these previous efforts, and finally defined a set of core future research pathways for GIS-based energy infrastructure planning, with a focus on the use of renewable energy sources. Tim Unterluggauer et al. [20] reviewed different methods and technologies for the planning of electric vehicle charging infrastructure and proposed various models and algorithms to address the location selection, scale determination, and charging demand prediction of charging stations. Yinan Liu et al. [21] gave an update on typical and innovative carbon capture and storage (CCS) technologies combined with solar energy, outlining thermodynamic processes, including chemical reactions, operating conditions, and efficiency indicators of the five typical technologies, in order to clarify the current level of development. From conventional coal-powered systems to renewable energy alternatives, from conventional oil storage to underground oil storage caverns, and from gas stations to electric vehicle charging devices, these research endeavors adopt different design philosophies and construction technologies yet share one unchanging objective [22–25]. They aim to architect and preserve efficient, reliable, and sustainable energy infrastructure systems, systems that exist symbiotically with the ecosystem and lead the charge in reducing our carbon footprint [26–29].

However, the complex terrain of a service environment, coupled with diverse and random loads, constitutes the primary obstacles preventing infrastructure engineers and researchers from materializing this objective [30,31]. While hybrid solar–wind–biomass energy systems have great potential for powering sustainable cities in the future, the complexity of their planning and design still limits their widespread implementation. By studying Vancouver, the researchers assessed the impact of economies of scale on hybrid renewable energy systems and found that the average cost of electricity (COE) of systems at different scales was slightly different [32–34]. Abdullahi Mohamed Samatar et al. [35] discussed the current situation and potential of solar energy utilization in Somalia, which, despite its abundant solar potential, still faces challenges such as a lack of infrastructure, high initial costs, and lack of energy awareness. The unpredictability of load demand, variability in energy sources, and intricacies of system design and maintenance all pose considerable challenges in achieving this goal [36,37]. Infrastructure engineers and researchers, therefore, must leverage technological advances, strategic planning, and innovative thinking to create more sustainable and efficient energy infrastructures [38,39].

The construction of energy infrastructure requires sufficient knowledge of geotechnical engineering, structural engineering, and the physical properties of energy materials, as well as the mechanical properties of the construction materials [40]. To enhance the service performance and service life of energy infrastructure, innovations in construction materials, structures, foundations, and construction measures are necessary [41–43]. Zaghloul et al. [44] conducted a study to determine the effect of fiber arrangement on the mechanical properties of glass-fiber-reinforced polymeric materials (GFRP). This paper focuses on the shrinkage factor, the tensile fatigue behavior of pultruded-glass-fiber-reinforced polyester, and its life estimation model. Cawas Phiroze Nazir [45] discussed the technical and economic feasibility of offshore hydropower plants using hydropower to develop marine energy reserves as renewable energy sources. Since the constantly charged stream provides the energy the plant uses to generate electricity, and the water is not used up in the process, it is a clean and renewable source of energy, the plant does not use fuel, and there is no uncertainty about transportation, storage, treatment costs, or fuel pricing, reducing the dependence on conventional fuels in other countries and being more environmentally friendly than traditional hydroelectric power plants [46,47]. Feng Shan et al. [48] introduced the performance evaluation and application of photovoltaic collectors and systems, analyzed and discussed the performance of photovoltaic hot-air collectors and photovoltaic hot water collectors, and introduced the application of photovoltaic thermal systems, such as building-integrated photovoltaic thermal systems, concentrated photovoltaic thermal systems, and photovoltaic heat pump systems. Jasem Alazemi et al. [49] discussed that

hydrogen stations are one of the most important parts of the power distribution infrastructure needed to support the operation of hydrogen fuel cell electric vehicles and hydrogen combustion engine vehicles. Up to this point, notwithstanding significant strides made in improving construction methodologies, developing sophisticated construction materials, and optimizing construction theories, a noticeable disconnect still exists [50–53]. Such a gap shifts between the specific requirements of sophisticated energy systems and the real-world conditions currently characterizing energy infrastructure. This indicates that developing advanced methods for effective and efficient technologies, equipment, and theoretical concepts to improve the design, construction, and maintenance of energy infrastructure remains a persistent challenge [54–57].

This Special Issue seeks to showcase and deliberate on recent studies, innovative methodologies, case studies, and review articles that describe the current status and notable advances within this research area. Submissions focusing on confronting current challenges associated with physical experiments, computational models, and theoretical analyses pertinent to the construction technology of energy infrastructure are welcomed. When feasible, such submissions should incorporate pertinent field observations from practice. Papers submitted on new and emerging topics within this discipline are also encouraged. Contributions in the form of theoretical papers are certainly welcomed, with a particular emphasis placed on submissions of practice-oriented papers and those based on computational methods. This Special Issue seeks contributions spanning a broad range of topics that are related, but not limited, to the following:

- Stability analysis of oil and gas pipelines/tunnels in multi-field coupling environments;
- Multi-hazards on energy infrastructure and lifeline structures;
- Reinforcement analysis of infrastructure and lifeline structures;
- Construction technology of renewable energy infrastructure;
- Hazard analysis of earthquakes and strong winds;
- Monitoring, spatial–temporal prediction modeling, and early warning of facility failures using advanced measurement systems;
- Sustainable and innovative materials for energy infrastructure;
- Construction technology and durability of nuclear waste treatment facilities under extreme conditions;
- New computational and experimental methods investigating the foundations of advanced energy infrastructures.

## 2. Outlook of This Special Issue

This Special Issue of *Energies* was successfully organized with the unwavering support from the editorial team of the journal and the MDPI publishing team. The processing time for the articles averaged approximately 39 days. The Guest Editor is very much grateful to all reviewers for reviewing and revising the manuscripts. Devoting their valuable time to reviewing papers is essential for upholding the voluntary peer review process and is highly commendable. Through their constructive feedback for authors and confidential reports to the editors, they have played a significant role in maintaining the high academic standard of this journal.

The response to our call for papers was excellent, with the following statistics:

- Submissions: (23);
- Publications: (10);
- Rejections: (13);

Article types: review articles (1); research articles (9).

The geographical distribution of authors (corresponding author and first author) by country for the published articles is presented in Table 1, in which it is possible to observe 20 authors from two countries. Note that it is usual for an item to be signed by more than one author and for authors to collaborate with others from different affiliations.

**Table 1.** Authors' geographical distribution.

| Country | Authors |
|---|---|
| China | 18 |
| Korea | 2 |

The corresponding authors and first authors of this Special Issue and their first affiliations are reflected in Table 2.

**Table 2.** Authors and affiliations.

| Author | First Affiliation | Country | Reference |
|---|---|---|---|
| Chaozhe Zhang | Southeast University | China | [58] |
| Jianyong Han | School of Civil Engineering, Shandong Jianzhu University | China | [58] |
| Li Tian | Shandong University | China | [58] |
| Duc-Vu Ngo | Kunsan National University | Korea | [59] |
| Dong-Hyawn Kim | Kunsan National University | Korea | [59] |
| Hong Yu | Electric Power Research Institute, Yunnan Power Grid Co., Ltd. | China | [60] |
| Hua-Dong Zheng | Wuhan University of Technology | China | [60] |
| Shidan Chi | Shandong Electric Power Engineering Consulting Institute Co., Ltd. | China | [61] |
| Tao Luan | Shandong University | China | [61] |
| Sifeng Zhang | Shandong Jianzhu University | China | [62] |
| Guojian Zhang | Shandong Jianzhu University | China | [62] |
| Hao Liu | China University of Geosciences | China | [63] |
| Meixia Zhang | China University of Geosciences | China | [63] |
| Yunjuan Chen | Shandong Jianzhu University | China | [64] |
| Tianshuo Xu | China University of Geosciences | China | [65] |
| Peng Zhang | China University of Geosciences | China | [65] |
| Junkuo Li | State Grid Hebei Economic Research Institute | China | [66] |
| Chuncheng Liu | Northeast Electric Power University | China | [66] |
| Peinan Li | Donghua University | China | [67] |
| Xi Wang | Tongji University | China | [67] |

Table 3 summarizes the research carried out by identifying the topics to which they belong, according to the proposed topics in this Special Issue. A clear dominance is observed in two research topics in this Special Issue, specifically "Pipeline" and "Transmission Tower".

**Table 3.** Research topics.

| Research Topics | Number of Publications | References |
|---|---|---|
| Low-carbon pile | 1 | [58] |
| Non-destructive test | 1 | [59] |
| Transmission tower | 2 | [60,66] |
| Grate biomass boiler | 1 | [61] |
| Pipeline | 3 | [62,63,65] |
| Hydraulic coupling | 1 | [64] |
| Special-shaped shield tunnel | 1 | [67] |

The summary of the articles published in this Special Issue is discussed in the subsequent sections of this editorial.

## 3. A Review of This Special Issue

A number of articles involving several subjects about the construction technology of advanced energy infrastructure have been published in this Special Issue. Their details and a brief summary of the addressed issues are provided in the following.

A Review of Efficient and Low-Carbon Pile Technologies for Extra-Thick Soft Strata by Chaozhe Zhang, Jianyong Han, Songyu Liu, Zhenglong Cao, Chen Jiang, Xuhan Diao, Guangwei Chen and Li Tian [58].

The authors discuss the challenges posed by limited bearing capacity in soft strata and the need for innovative pile technologies. They focus on composite pile technologies that combine rigid piles with flexible columns, such as the stiffened deep mixing (SDM) column, squeezed branch pile, pre-bored grouting plated nodular (PGPN) pile, precast cement pile reinforced by cemented soil with a variable section (PCCV), and carbonized composite pile (CCP). The authors also review low-carbon energy-efficient curing technologies that have been developed for geotechnical engineering. This paper provides insights into efficient and low-carbon pile technologies and offers guidance for industry practitioners.

Reliability Analysis and Life Prediction of Aging LNG Unloading Arms based on Non-Destructive Test Data by Duc-Vu Ngo, Jong-Kwon Lim and Dong-Hyawn Kim [59].

This study examines unloading arms (ULAs) that have been in service for over 30 years and analyzes their structural deterioration and integrity. The authors use a finite-element program to model the aging ULAs and update the thickness dimensions of structural members based on NDT data. A reliability assessment is then conducted using the stress distribution of the main structural components under external loads, specifically wind speed. This study also constructs a time-dependent reliability index curve to predict the failure probability of the components during their remaining lifetime. The findings suggest that the current condition of the ULA system is satisfactory for current loading conditions.

Failure Patterns of Transmission Tower-Line System Caused by Landslide Events by Hong Yu, Hao Li, Zhi-Qiang Zhang, Gui-Feng Zhang, Da-Hai Wang and Hua-Dong Zheng [60].

This article investigates the collapse patterns of transmission tower-line structures under landslides. This study aims to understand the failure mechanisms of these structures and the impact of landslides on their collapse patterns. This research uses the explicit analysis method to analyze the nonlinear dynamic response equations of the structure. Two foundation failure cases under landslides are considered, and the displacement responses and progressive collapse behavior are analyzed. This study finds that the failure patterns of the tower-line structure under landslides are influenced by the failure mode of the tower foundation and the pulling effect of cables.

Numerical Simulation of Tail Over-Fire Air Supply of a Grate Biomass Boiler by Shidan Chi, Yan Liang, Weixi Chen, Zhen Hou and Tao Luan [61].

This article explores the effects of tail burnout air's incidence angle, wind speed, and pipe diameter on the flow field distribution and movement of unburned carbon particles in a grate biomass boiler. This study uses ANSYS software to simulate these effects and obtain the influence rules of the tail burnout air parameters on combustion and particle movement. The results show that the setting of burnout air at the tail of the grate can improve air flow field organization, change particle distribution, and prolong particle residence time on the grate. Increasing the outlet velocity and pipe diameter of the tail burnout air leads to an increased movement and burnout of grate particles, resulting in improved boiler efficiency.

Seepage Performance of Fibre Bundle Drainage Pipes: Particle Flow Simulation and Laboratory Testing by Sifeng Zhang, Guozhang Ren, Guojian Zhang, Ziyin Ren, Chong Xia and Yuan Gao [62].

This article explores the seepage performance of fiber bundle drainage pipes in slope engineering projects. The authors conducted particle flow simulations and laboratory tests to investigate the influences of soil particles on the clogging of geotextile filters and drainage pipes, as well as the seepage rates. They found that higher water pressure, smaller flower hole intervals in the pipe, greater soil friction angles, and smaller pipe inclination angles are

less conducive to drainage. They also discovered that installing a fiber bundle in drainage pipes can improve seepage and drainage performance under silting conditions. The authors tested five types of fiber bundles and found that plastic rope provided the best drainage effect. The arrangement of fiber bundles also affected drainage, with uneven arrangements being more effective with plastic rope and cotton rope, and uniform arrangements being more effective with nylon rope, hemp rope, and polyester rope.

Properties of lightweight Controlled Low-Strength Materials using construction waste and EPS for Oil and Gas Pipelines by Hao Liu, Kaizhi Liu, Yiqi Xiao, Peng Zhang, Meixia Zhang, Youzeng Zhu, Kaixin Liu, Tianshuo Xu and Rui Huang [63].

This study investigates the use of construction waste and expanded polystyrene (EPS) as fine aggregate to prepare lightweight controlled low-strength materials (CLSMs) for oil and gas pipelines. The EPS is modified with ethylene vinyl acetate-resin (EVA) to improve its compatibility with cementitious materials. The CLSM's fluidity, strength, and microstructure are studied through experiments and microscopic analysis. The results show that the surface modification of EPS with EVA improves its compatibility with cementitious materials and allows for a fluidity greater than 200 mm. The volume ratio of EPS to construction waste and the cement content significantly affect the material's strength development. This study suggests that the EPS content should not be too large to achieve the desired mechanical properties. Microscopic analysis reveals that an increase in EPS content leads to poor pore uniformity and a loose mesh structure, which hinders strength development.

A study on the influence of joint locations and hydraulic coupling actions on rock masses' failure process was conducted by Yunjuan Chen, Tao Gao, Fuqiang Yin, Xiaozhi Liu, and Jun Wang [64].

This article explores the distribution of joints and fissures in rock masses and their impact on the stability of surrounding rock in underground engineering construction. The authors compare and analyze two simulation programs, Discontinuous Deformation Analysis for Rock Failure (DDARF) and Rock Failure Process Analysis (RFPA), based on laboratory tests. They then use RFPA to analyze the failure process, stress state, acoustic emission characteristics, and energy dissipation laws of rock masses with different joint locations under hydraulic coupling conditions. The results show that the location of joints affects the failure process, with a large tensile stress region generated on both ends of the original joint and the damage effect angle increasing gradually from the middle joint to the marginal joint. The water pressure in cracks promotes the generation of tensile stress, and its effect on crack failure depends on the proximity of the joint to the middle part of the specimen.

Key techniques for gas pipelines rapid jacking and lying: A case study in project of 'Jingshihan' gas pipeline in China by Tianshuo Xu, Le Wang, Peng Zhang, Yuheng Zhou, Kaixin Liu, Xin Feng, Yongchun Qi and Cong Zeng [65].

This article discusses the key techniques for the rapid jacking and laying of pipelines, using the case study of the 'Jingshihan' gas pipelines in China. The authors highlight the need to improve the construction process of oil and gas pipelines due to the rapid growth of the industry. They introduce the direct pipe laying method, which combines the advantages of horizontal directional drilling and pipe jacking construction. The method is suitable for pipeline crossings in different strata and offers advantages such as less construction land, high speed, and reversibility. The authors investigate the application status of the direct pipe laying method in the 'Jingshihan' gas pipeline project and discuss various aspects of the construction process, including the working well, equipment selection, guiding control technology, supporting equipment installation, and drag reduction measures. They also analyze the factors influencing thrust force and trajectory deviation.

Collapse Mechanism of Transmission Tower Subjected to Strong Wind Load and Dynamic Response of Tower-line System by Junkuo Li, Fan Gao, Lihuan Wang, Yaning Ren, Chuncheng Liu, Aiquan Yang, Zhao Yan, Tao Jiang and Chengbo Li [66].

This article investigates the collapse mechanism and failure modes of transmission towers under strong wind loads. The authors use the finite element method (FEM) to analyze the ultimate bearing capacity of a typical 220 kV transmission tower. The results show that the collapse of the tower under strong wind loads is usually due to the buckling of the leg members. The authors also establish a finite element model to analyze the dynamic response of the tower-line system under fluctuating wind. The results show that the wind-induced coefficients designed by current codes accurately assess the collapse displacement of the transmission tower, but the load-carrying capacity of the tower in the plastic phase can be overestimated by static calculation results.

A Study of the Segment Assembly Error and Quality Control Standard of Special-Shaped Shield Tunnels by Peinan Li, Zeyu Dai, Xi Wang, Jun Liu, Yi Rui, Xiaojun Li, Jie Fan and Peixin Chen [67].

This study focuses on the segment assembly error and quality control standard of specially shaped shield tunnels. The researchers investigated the formation mechanism and control measures of lining segment assembly defects in a quasi-rectangular shield tunnel project. They developed a simulation program to calculate the error in segment assembly by quantifying the manufacturing error and positioning error. They also established a finite element model to analyze the mechanical behavior of key blocks and proposed a control standard for the key block assembly error. The results showed that the assembly quality can be improved by assembling the LZ block first and applying error control measures.

## 4. Conclusions

The contributions in this Special Issue discussed a wide range of subjects relevant to the construction technology and applications of advanced energy infrastructure. Construction technologies are crucial for the update and development of energy infrastructure. Research on advanced construction technology and energy infrastructure theory is therefore very important for the energy industry. Even though the articles reported very interesting applications and relative techniques developed thus far, there are still many gaps to fill to satisfy the rapidly increasing demand for energy utilization.

Combined with the ever-increasing global demand for energy, it has become essential to find new solutions, based on the background of the complex service conditions of energy infrastructure and the development of technology of renewable energy resources. The expansion of new energy sources, particularly in the realm of electric vehicles, has underscored the significance of robust charging infrastructure. Simultaneously, the impact of conventional power facilities on the environment remains a pivotal concern. Traditional electricity production and transmission, often reliant on fossil fuels, have contributed significantly to air pollution and greenhouse gas emissions. However, the steady advancement of renewable energy sources, including wind and solar power, has provided a means to mitigate environmental degradation. By enhancing power infrastructure and embracing greener energy production methods, substantial progress can be made in minimizing adverse environmental effects.

Addressing the security of energy infrastructure also holds paramount importance, which includes two primary issues: structural safety and network security. Regardless of whether it is the collapse of transmission towers or hydroelectric dam failures, incessant issues regarding the structural stability of energy infrastructure bring vast damages to the energy industry and national economy. To address these structural safety issues, further advancements in the technology and theory of infrastructure construction should be enhanced, and they should propel the development of associated construction standards for energy infrastructure. On the other hand, the evolution of interconnected systems, from charging stations to intelligent grids, accentuates the need for a holistic security approach encompassing both physical and cyber safeguards. Governments and industry stakeholders must collaborate to establish robust standards, protocols, and contingency plans to counter potential threats. This comprehensive strategy entails safeguarding not

only physical assets but also bolstering cybersecurity measures to thwart cyberattacks and prevent data breaches.

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